

1c_bonus-D.md

Ready to Suffer? –here’s the next realism bump for Lab 1C-Bonus-D:

- 1) Zone apex (

Lab 1C Bonus D: Execution Plan

Overview

This bonus adds two critical production features:

1. **Apex domain** (`williebright.com`) pointing to your ALB (so users don't need to type `app.`)
2. **ALB access logs** to S3 for incident response and audit trails

Step 1: Add Variables to `03-variables.tf`

Append these to the bottom of your existing variables file:

```
# =====
# BONUS D: ALB Access Logging Variables
# =====
variable "enable_alb_access_logs" {
  description = "Enable ALB access logging to S3."
  type        = bool
  default     = true
}
variable "alb_access_logs_prefix" {
  description = "S3 prefix for ALB access logs."
  type        = string
  default     = "alb-access-logs"
}
```

Step 2: Create New File ``04-1cd-logging-apex.tf``

This file creates the S3 bucket for logs and the apex A record.

```
#####
# BONUS D: ALB Access Logs S3 Bucket + Apex Record
#####
# =====
# S3 Bucket for ALB Access Logs
# =====
# Explanation: Chewbacca keeps flight logs—this bucket stores
ALB access logs for audits and incident response.
resource "aws_s3_bucket" "helga_alb_logs_bucket01" {
  count  = var.enable_alb_access_logs ? 1 : 0
  bucket = "${var.project_name}-alb-logs-${
```

Step 2.5: Fix Subnet References (Count-Based Resources)

Problem Encountered:

When referencing `aws_subnet.helga_public_subnets` in the ALB resource, Terraform threw this error:

```
| Error: Missing resource instance key
|
|   on
```

Root Cause:

The subnet resource uses `count`, which creates multiple instances. You can't reference `.id` directly—you must specify which instance using an index.

Fix Applied:

Updated the `subnets` attribute to use explicit indices:

```
subnets = [aws_subnet.helga_public_subnets[0].id, aws_subnet.helga_public_subnets[1].id]
```

Alternative (Splat Expression):

For dynamic scaling, use the splat operator to automatically expand all instances:

```
subnets = aws_subnet.helga_public_subnets[*].id
```

Step 3: Modify Existing ALB Resource in `04-1cb-Main.tf`

Find your `aws_lb.helga_alb01` resource and **add** the `access_logs` block inside it:

```
resource "aws_lb" "helga_alb01" {
  name           = "${var.project_name}-alb01"
  internal       = false
  load_balancer_type = "application"
  security_groups = [aws_security_group.helga_alb_
```

Alternative (static block):

```
access_logs {
  bucket = var.enable_alb_access_logs ? aws_s3_bucket.helga_alb_logs_bucket01[0].bucket : ""
  prefix = var.alb_access_logs_prefix
  enabled = var.enable_alb_access_logs
}
```

Step 4: Add Outputs to `05-outputs.tf`

Append these to your outputs file:

```
# =====  
# BONUS D: Apex + Logging Outputs  
# =====  
output "helga_apex_url_https" {  
  value = "
```

Step 5: Apply

```
terraform init  
terraform plan  
terraform apply
```

Step 6: CLI Verification

1. Verify Apex Record Exists

```
aws route53 list-resource-record-sets \  
  --hosted-zone-id <hosted zone ID> \  
  --query "ResourceRecordSets[?Name=='
```

2. Verify ALB Logging is Enabled

First, get your ALB ARN:

```
aws elbv2 describe-load-balancers \  
  --names helga-alb01 \  
  --query "LoadBalancers[0].LoadBalancerArn" --output text
```

Then check the attributes:

```
aws elbv2 describe-load-balancer-attributes \  
  --load-balancer-arn arn:aws:elasticloadbalancing:us-east-2:  
919113286081:loadbalancer/app/helga-alb01/4cc4f3a53980daa6 \  
  --query "Attributes[0].Value" --output text
```

```
--query "Attributes[?Key=='access_logs.s3.enabled' || Key==  
'access_logs.s3.bucket' || Key=='access_logs.s3.prefix']"
```

Expected:

- `access_logs.s3.enabled` = `true`
- `access_logs.s3.bucket` = your bucket name
- `access_logs.s3.prefix` = `alb-access-logs`

2.5: Subnet Reference Fix Encountered Here

Issue: During verification, if ALB logging shows `access_logs.s3.enabled = false`, trace back to the ALB resource. A common blocker is the subnet reference error:

```
| Error: Missing resource instance key  
|   subnets = [aws_subnet.helga_public_
```

Fix: Use explicit indices or the splat operator:

```
# Explicit  
subnets = [aws_subnet.helga_public_subnets[0].id, aws_subnet.  
helga_public_subnets[1].id]  
  
# Or splat (dynamic)  
subnets = aws_subnet.helga_public_subnets[*].id
```

After fixing, re-run `terraform apply` and verify again.

3. Generate Traffic

```
curl -I
```

3.5: Certificate Mismatch Fix Encountered Here

Issue: After fixing DNS/nameserver mismatch, `curl -I https://williebright.com` returned:

```
curl: (60) schannel: SNI or certificate check failed: SEC_E_W  
RONG_PRINCIPAL (0x80090322)
```

Root Cause: The ACM certificate only covered `app.williebright.com`, not the apex `williebright.com`. When hitting the apex domain, the TLS handshake failed because the cert didn't match.

Diagnosis: Check what domains your cert covers:

```
aws acm describe-certificate \  
  --certificate-arn <YOUR_CERT_ARN> \  
  --query "Certificate.SubjectAlternativeNames"
```

Fix Applied: Updated the ACM certificate to include both domains:

```
resource "aws_acm_certificate" "helga_cert01" {  
  domain_name      = "
```

Then updated the HTTPS listener to use the new certificate:

```
resource "aws_lb_listener" "helga_https_listener01" {  
  load_balancer_arn = aws_lb.helga_alb01.arn  
  port              = 443  
  protocol          = "HTTPS"  
  ssl_policy        = "ELBSecurityPolicy-TLS13-1-2-2021-06"  
  certificate_arn    = aws_acm_certificate.helga_cert01.arn  
  
  default_action {  
    type           = "forward"  
    target_group_arn = aws_lb_target_group.helga_tg01.arn  
  }  
}
```

After `terraform apply`, both domains returned `HTTP/1.1 200 OK`.

4. Verify Logs Arrived in S3 (Wait 5-10 Minutes)

```
aws s3 ls s3://<BUCKET_NAME>/alb-access-logs/AWSLogs/919113286081/elasticloadbalancing/ --recursive | head
```

```
aws s3 ls s3://helga-alb-logs-919113286081/alb-access-logs/AWSLogs/919113286081/elasticloadbalancing/ --recursive | head
```

Troubleshooting

Issue	Likely Cause	Fix
S3 bucket policy error	Bucket policy doesn't allow ELB service account	Verify the <code>aws_s3_bucket_policy</code> resource has the correct ELB service account ARN for your region
No logs appearing	ALB hasn't received traffic yet OR logs take 5-10 min	Generate traffic with <code>curl</code> , wait, check again
Apex domain not resolving	DNS propagation	Wait a few minutes, run <code>nslookup williebright.com</code>
Certificate error on apex	ACM cert doesn't include apex domain	Ensure your ACM certificate includes both <code>app.williebright.com</code> AND <code>williebright.com</code>

Important Note: ACM Certificate for Apex

If your current ACM certificate only covers `app.williebright.com`, you'll get a certificate error when hitting `https://williebright.com`. You need to either:

1. **Add apex to existing cert** — Request a new certificate that includes both `williebright.com` AND `app.williebright.com`
2. **Use a wildcard** — Request a certificate for `*****.williebright.com` AND `williebright.com`

Check your current cert:

```
aws acm describe-certificate \
  --certificate-arn arn:aws:acm:us-east-2:919113286081:certificate/9abc064c-fee6-4a47-b6d5-756bc8bf6d47 \
  --query "Certificate.SubjectAlternativeNames"
```

Career Takeaway

ALB access logs are **incident response fuel**. When something breaks at 2 AM, these logs tell you:

- Which client IPs are hitting your ALB
- What paths they're requesting
- Response codes (4xx, 5xx patterns)
- Backend target behavior
- Request latency

Combined with WAF logs and CloudWatch alarms, you can answer: *"Is this attackers, misroutes, or a downstream failure?"*

✅ Bonus D Complete — Summary

Completed: January 17, 2026

Objectives Achieved

Objective	Status
Apex domain (williebright.com) → ALB	✅ Complete
ALB access logs → S3	✅ Complete
Both domains returning 200 OK	✅ Verified
Logs appearing in S3 bucket	✅ Confirmed

Issues Encountered & Resolved

1. **Subnet Reference Error** — Fixed by using explicit indices `[0]` and `[1]` for count-based resources
2. **Nameserver Mismatch** — Route53 hosted zone nameservers didn't match registered domain; synced via Terraform
3. **Certificate Mismatch** — ACM cert only covered `app.williebright.com`; updated to include apex `williebright.com` and refreshed HTTPS listener

Final Verification Output

```
aws s3 ls s3://helga-alb-logs-919113286081/alb-access-logs/AW
SLogs/919113286081/elasticloadbalancing/ --recursive | head

2026-01-16 22:45:02 209 alb-access-logs/.../20260117T0345Z_1
8.216.38.254_sjewsm9p.log.gz
2026-01-16 22:50:02 1711 alb-access-logs/.../20260117T0350Z_1
8.216.38.254_2ostge9o.log.gz
2026-01-16 22:50:02 1649 alb-access-logs/.../20260117T0350Z_
3.23.25.171_5rg86jcn.log.gz
...
```

Result: Logs flowing every 5 minutes from ALB to S3. Production-grade observability achieved.

```
brigh@DESKTOP-62DEPP9 MINGW64 ~/Documents/TheoWAF/class7/AWS/Classes/armageddon_3 (master)
$ aws s3 ls s3://helga-alb-logs-919113286081/alb-access-logs/AWSLogs/919113286081/elasticloadbalancing/ --re
cursive | head
7/919113286081_elasticloadbalancing_us-east-2_app.helga-alb01.4cc4f3a53980daa6_20260117T0350Z_3.23.25.171_5r
g86jcn.log.gz
2026-01-16 22:55:02      210 alb-access-logs/AWSLogs/919113286081/elasticloadbalancing/us-east-2/2026/01/1
7/919113286081_elasticloadbalancing_us-east-2_app.helga-alb01.4cc4f3a53980daa6_20260117T0355Z_18.216.38.254_
62y56ibr.log.gz
2026-01-16 22:55:02     1062 alb-access-logs/AWSLogs/919113286081/elasticloadbalancing/us-east-2/2026/01/1
7/919113286081_elasticloadbalancing_us-east-2_app.helga-alb01.4cc4f3a53980daa6_20260117T0355Z_3.23.25.171_60
afud27.log.gz
2026-01-16 23:00:02     1191 alb-access-logs/AWSLogs/919113286081/elasticloadbalancing/us-east-2/2026/01/1
7/919113286081_elasticloadbalancing_us-east-2_app.helga-alb01.4cc4f3a53980daa6_20260117T0400Z_18.216.38.254_
2zeyewyu.log.gz
2026-01-16 23:00:02      596 alb-access-logs/AWSLogs/919113286081/elasticloadbalancing/us-east-2/2026/01/1
7/919113286081_elasticloadbalancing_us-east-2_app.helga-alb01.4cc4f3a53980daa6_20260117T0400Z_3.23.25.171_11
cpyrmm.log.gz
2026-01-16 23:05:02      519 alb-access-logs/AWSLogs/919113286081/elasticloadbalancing/us-east-2/2026/01/1
7/919113286081_elasticloadbalancing_us-east-2_app.helga-alb01.4cc4f3a53980daa6_20260117T0405Z_18.216.38.254_
3ex93iyg.log.gz
2026-01-16 23:10:02    11668 alb-access-logs/AWSLogs/919113286081/elasticloadbalancing/us-east-2/2026/01/1
7/919113286081_elasticloadbalancing_us-east-2_app.helga-alb01.4cc4f3a53980daa6_20260117T0410Z_18.216.38.254_
4njees9h.log.gz
2026-01-16 23:15:02    1491 alb-access-logs/AWSLogs/919113286081/elasticloadbalancing/us-east-2/2026/01/1
7/919113286081_elasticloadbalancing_us-east-2_app.helga-alb01.4cc4f3a53980daa6_20260117T0415Z_18.216.38.254_
hsroqzm.log.gz
```

```
brigh@DESKTOP-62DEPP9 MINGW64 ~/Documents/TheoWAF/class7/AWS/Classes/armageddon_3 (master)
$ aws elbv2 describe-load-balancer-attributes \
  --load-balancer-arn arn:aws:elasticloadbalancing:us-east-2:919113286081:loadbalancer/app/helga-alb01/4cc4f
3a53980daa6 \
  --query "Attributes[?Key=='access_logs.s3.enabled' || Key=='access_logs.s3.bucket' || Key=='access_logs.s3
.prefix']"
[
  {
    "Key": "access_logs.s3.enabled",
    "Value": "true"
  },
  {
    "Key": "access_logs.s3.bucket",
    "Value": "helga-alb-logs-919113286081"
  },
  {
    "Key": "access_logs.s3.prefix",
    "Value": "alb-access-logs"
  }
]
```

```

    },
    {
      "Value": "ns-960.awsdns-56.net."
    },
    {
      "Value": "ns-1414.awsdns-48.org."
    },
    {
      "Value": "ns-1580.awsdns-05.co.uk."
    }
  ]
},
{
  "Name": "williebright.com.",
  "Type": "SOA",
  "TTL": 900,
  "ResourceRecords": [
    {
      "Value": "ns-269.awsdns-33.com. awsdns-hostmaster.amazon.com. 1 7200 900 1209600 86400"
    }
  ]
}
]

```

```

brigh@DESKTOP-62DEPP9 MINGW64 ~/Documents/TheoWAF/class7/AWS/Classes/armageddon_3 (master)
$ aws route53 list-resource-record-sets \
  --hosted-zone-id Z00908063NUR2ECTZKXUK \
  --query "ResourceRecordSets[?Name=='williebright.com.']"
[
  {
    "Name": "williebright.com.",
    "Type": "A",
    "AliasTarget": {
      "HostedZoneId": "Z3AADJGX6KTTL2",
      "DNSName": "helga-alb01-1017232551.us-east-2.elb.amazonaws.com.",
      "EvaluateTargetHealth": true
    }
  },
  {
    "Name": "williebright.com.",
    "Type": "NS",
    "TTL": 172800,
    "ResourceRecords": [
      {
        "Value": "ns-269.awsdns-33.com."
      }
    ]
  }
]

```